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# A VOLUME-LIMITED SEARCH FOR TECHNOSIGNATURES WITHIN 20 PC USING THE PUBLIC ALMA ARCHIVE

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## ABSTRACT

We present the motivation, design, methodology and first results of a volume-limited search for radio technosignatures among all stars within 20 pc of the Sun that have public observations in the Atacama Large Millimeter/submillimeter Array (ALMA) science archive. Unlike the great majority of technosignature surveys conducted to date, which are either wide, shallow all-sky surveys or targeted campaigns preferentially selecting exoplanet hosts, young stellar objects, or other astrophysically "interesting" systems, our sample of 120 stars is constructed to be representative of the general stellar population in the immediate solar neighbourhood, irrespective of spectral type, age, multiplicity, or known planetary or circumstellar-disc status. We mine the ALMA archive opportunistically: rather than requesting dedicated observing time, we reprocess existing calibrated (and, where necessary, re-calibrated) visibility data to search simultaneously for narrowband Doppler-drifting technosignature candidates and for anomalous broad-band continuum emission, at frequencies ($\sim$90–950 GHz) that remain almost entirely unexplored by the technosignature literature. We supplement the classical narrowband/continuum search with two further analyses that exploit properties unique to this survey and unavailable to any single-dish technosignature programme: a closure-phase point-source vetting test for any candidate that crosses threshold, and a population-level, spectral-type-normalised continuum anomaly check made possible by the sample's volume-limited, unbiased design. As of this version, 19 of the planned 120 target/bands have been searched to completion. No credible technosignature has been found – one narrowband candidate initially flagged by the automated pipeline is identified as the CO(1–0) transition and is not of technosignature origin – and we report EIRP upper limits in the range $1.6 \times 10^{13}$–$2.8 \times 10^{15}$ W (median $2.1 \times 10^{14}$ W) and five continuum detections (one an unresolved close binary), alongside fourteen deep non-detections reaching a median $5\sigma$ continuum limit of $\sim$0.6 mJy. The survey is ongoing; this paper will be updated as further targets are completed.

*Subject headings:* extraterrestrial intelligence — radio lines: stars — submillimetre: stars — surveys — astrobiology

## 1. INTRODUCTION

The search for extraterrestrial intelligence (SETI) rests on the premise that a technological civilisation, whether deliberately signalling or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Since the foundational proposal of Cocconi & Morrison (1959) that microwave frequencies near the 21 cm hydrogen line would be an obvious "meeting place" for interstellar communication, and Frank Drake's first observational test of the idea with Project Ozma (Drake 1960), the radio band has remained the dominant search space for technosignatures, for reasons that are both historical and physically well motivated: radio photons are cheap to produce in enormous numbers, propagate through the interstellar medium and planetary atmospheres with little attenuation, and (for a sufficiently narrowband signal) can in principle be distinguished from any known natural astrophysical process by their spectral compactness and Doppler dynamics (Sheikh 2020).

Six decades on, radio technosignature searches have grown from single-target, single-frequency pointings into surveys of many thousands of stars across broad swathes of the radio spectrum. Yet, as Wright et al. (2018) demonstrated quantitatively with their "Cosmic Haystack" formalism, even the combined efforts of every major SETI programme to date have sampled only an infinitesimal fraction of the plausible multi-dimensional parameter space of transmitter frequency, bandwidth, power, location, and duty cycle. Persistent, non-trivial gaps remain: almost the entire millimetre and submillimetre regime above $\sim$50 GHz is essentially unsearched; most surveys have targeted specific classes of star (nearby exoplanet hosts, young stellar objects with protoplanetary discs, stars in the "Earth Transit Zone") rather than an unbiased, complete sample of the local stellar population; and only a small number of studies have systematically reprocessed *existing* interferometric archives for technosignatures, as opposed to requesting dedicated telescope time.

This paper describes a new survey designed to address several of these gaps simultaneously. We construct a volume-limited sample of every star within 20 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archival data for both narrowband technosignature candidates and anomalous continuum emission, applying corrections for stellar proper motion and for the range of Doppler drift rates

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plausible for a transmitter on a planet in orbit around its host star. We further supplement this classical search with two analyses that exploit properties specific to this survey and unavailable to prior technosignature work: interferometric closure-phase vetting of any candidate signal, and a population-level anomaly check enabled by the sample's unbiased, volume-limited design. Section 2 reviews the relevant literature: the history of radio SETI (§2.1), the current generation of large-scale technosignature surveys (§2.2), the very limited body of work at millimetre/submillimetre frequencies (§2.3), the growing use of machine learning and anomaly detection in this field (§2.4), the statistical frameworks used to interpret survey non-detections (§2.5), the case for a volume-limited, representative sample (§2.6), and the prospects offered by next-generation facilities (§2.7). Section 3 summarises our sample selection, Section 4 our data-reduction and search methodology (including the two new analyses), Section 5 presents our results to date, and Section 6 discusses their interpretation and the survey's prospects. Full per-target tables are deferred to Appendix A so as not to interrupt the main narrative with tabular material.

## 2. BACKGROUND

### 2.1. *Six decades of radio SETI: a brief history*

Project Ozma (Drake 1960) observed two nearby Sun-like stars, $\tau$ Ceti and $\epsilon$ Eridani, with the 26 m Tatel telescope at Green Bank at 1420 MHz for a total of some 150 hours, and found no signal, but established both the methodology and the symbolic starting point for the modern field. Through the 1960s–1980s a steady stream of targeted and sky-survey searches followed, including Ozma II (Drake 1986), the Ohio State "Big Ear" survey (source of the famous "Wow!" signal), and the META and BETA all-sky surveys conducted from Harvard/Smithsonian facilities. Project Phoenix (Backus & Project Phoenix Team 2002), run by the SETI Institute from 1995–2004, was the most sensitive targeted survey of its era, observing on the order of 800 nearby Sun-like stars using the Arecibo, Parkes and Green Bank telescopes, and pioneered a rigorous methodology for distinguishing candidate signals from terrestrial radio-frequency interference (RFI) through simultaneous multi-site or multi-antenna verification. The Allen Telescope Array (ATA; Tarter 2001), jointly developed by the SETI Institute and UC Berkeley and first commissioned in 2007, was the first radio telescope purpose-built from the outset for commensal, wide-band SETI observing, and remains in active use, including in recent technosignature searches of interstellar objects (Sheikh et al. 2025b) and TRAPPIST-1 (Tusay et al. 2024). The distributed-computing project SETI@home (Von Korff et al. 2025) processed public-participant-donated CPU cycles to search Arecibo sky-survey data for narrowband and pulsed signals for two decades before its 2020 shutdown, and its final data-analysis summary was only published in 2025.

Two complementary observing strategies have been used throughout this history: *sky surveys*, covering a large fraction of the celestial sphere at modest sensitivity per pointing, and *targeted searches*, concentrating integration time on a preselected list of promising stars (e.g. Drake 1974). Both strategies persist in the modern era, but – as we discuss in §2.6 – the majority of targeted programmes have selected their samples on some presumed indicator of habitability or interest (proximity, Sun-like spectral type, known planets, or membership of the Earth Transit Zone), rather than aiming for a complete, unbiased census of the local stellar population.

### 2.2. *Modern large-scale radio technosignature surveys*

The current generation of technosignature science is dominated by the Breakthrough Listen initiative, launched in 2015 with a ten-year, \$100 million commitment to survey the one million nearest stars, the full Galactic plane and centre, the 100 nearest galaxies, and a sample of exoplanet and planet-candidate hosts, primarily using the Robert C. Byrd Green Bank Telescope (GBT) and the CSIRO Parkes "Murriyang" telescope, and now increasingly via commensal observing on MeerKAT (Isaacson et al. 2017). Its first major published dataset, Enriquez et al. (2017), searched 692 nearby stars over 1.1–1.9 GHz with the GBT and set limits on narrowband transmitters of EIRP $> 10^{13}$ W; this was followed by Price et al. (2020) (1327 stars, 1.10–3.45 GHz), Gajjar et al. (2019)-style searches of the restricted Earth Transit Zone at 3.95–8.00 GHz, and, most recently, a substantial expansion of automated, commensal technosignature searching with MeerKAT that has processed some 1.5 million coherent beams between 2023 and 2026 (Czech et al. 2026). In parallel, Margot et al. (2023) conducted a large citizen-science-style search of 11 680 stars with the GBT at 1.15–1.73 GHz, explicitly developing a statistical formalism for interpreting non-detections in terms of the fraction of stars that could host a detectable transmitter, *provided the observed sample is representative of the underlying stellar population of the search volume* – a condition that, as we discuss below, is often not satisfied by targeted, interest-selected samples.

Beyond Breakthrough Listen, the last five years have seen international radio facilities extend technosignature searches to new frequency regimes and observational strategies: the Five-hundred-metre Aperture Spherical Telescope (FAST; Zhang et al. 2020; Li et al. 2026) in China has conducted commensal searches with unmatched raw sensitivity at L-band, including recent narrowband searches of interstellar objects; LOFAR and the pathfinder NenuFAR have pushed technosignature searches down to 10–190 MHz, with LOFAR alone having surveyed over 1.6 million stars in commensal mode and dual-site LOFAR configurations enabling near-real-time RFI rejection (Johnson et al. 2023); the Sardinia Radio Telescope has conducted one of the first dedicated high-frequency ($> 18$ GHz) surveys of nearby stars and the Galactic Centre (Valente et al. 2025); and commensal technosignature back-ends are now operating or in development at the Karl G. Jansky Very Large Array (the COSMIC system; Tremblay et al. 2024), MeerKAT (Czech et al. 2021), and the Murchison Widefield Array (Morrison et al. 2023). Occultation-based strategies – observing a star during a planet-planet or planet-star occultation, when geometric alignment transiently favours detection of leakage or deliberate signals – have also grown in popularity, exemplified by recent Breakthrough Listen searches of TESS-selected eclipsing systems (Tusay et al. 2025).

### 2.3. *Millimetre and submillimetre technosignature searches*

In stark contrast to the radio regime below $\sim 10$ GHz, the millimetre and submillimetre bands remain almost entirely unexplored for technosignatures. Mason et al. (2024) conducted, to our knowledge, the first technosignature search ever performed with ALMA, searching archival Band 3 data (spectral windows centred at 90.642 and 93.151 GHz) for narrowband signals towards 28 Galactic stars selected from Gaia DR3. Their approach used a "stellar bycatch" method: rather than targeting stars directly, they searched for serendipitous stars falling within the undistorted field of view of four bright ALMA phase calibrators, since these calibrators are observed frequently and are not generally themselves of astrophysical interest. For the nearest star in their sample they placed a limit of $\mathrm{EIRP}_{\mathrm{min}} > 7 \times 10^{17}$ W, and discussed in detail the specific challenges of high-frequency technosignature work: substantially larger Doppler drift rates at fixed line-of-sight acceleration (since drift rate scales linearly with observing frequency), the resulting sensitivity loss ("smearing") for a fixed spectral and temporal resolution unless a drift-rate search is performed, and an increased incidence of spectral confusion from the dense molecular line forest present in protoplanetary discs and star-forming regions at these frequencies. Tusay et al. (2024) extended technosignature work on the TRAPPIST-1 system to the Allen Telescope Array, underlining the continued observational interest in this particular target but also its status as one of very few systems that has now been searched for technosignatures across a wide range of frequencies. Our own earlier work (White & Dey, in prep.; archived at `https://github.com/Tilanthi/SETI`) reproduced and extended the Mason et al. (2024) methodology directly on ALMA science-target (rather than calibrator-bycatch) fields for TRAPPIST-1 across Bands 3, 6 and 7, finding no detection and placing limits of $\mathrm{EIRP}_{\mathrm{min}} \sim 5\text{–}10 \times 10^{13}$ W, and subsequently built a ranked target list and pilot survey of 100 nearby, ALMA-observed stars, from which the present, volume-limited survey has grown. Beyond these efforts, we are not aware of any other published, systematic technosignature search using ALMA or any comparably sensitive millimetre/submillimetre interferometer, representing a substantial and readily addressable gap in frequency coverage relative to the sub-10 GHz literature.

The near-complete absence of mm/submm technosignature work to date is attributable less to any fundamental unsuitability of these frequencies – Sheikh's "nine axes of merit" framework (Sheikh 2020) notes that higher frequencies suffer less from interstellar scintillation and dispersion, and offer access to much larger instantaneous bandwidths – than to historical circumstance: sensitive mm/submm interferometers with the necessary spectral resolution and archival depth (chiefly ALMA) have only existed, and only accumulated a sufficiently large calibrated science archive, within roughly the last decade. This makes archival reprocessing, rather than dedicated proposal-driven observing, a particularly natural strategy in this regime: ALMA's archive already contains many thousands of hours of high-spectral-resolution observations of individual, nearby stars, taken for entirely unrelated astrophysical purposes (disc structure, molecular chemistry, stellar activity, debris-disc detection, calibration monitoring), each of which can in principle be re-searched for technosignatures at negligible additional observing cost.

### 2.4. *Machine learning and anomaly detection in technosignature searches*

The scale of modern technosignature datasets – hundreds of hours of high-time- and frequency-resolution data per survey, often containing many thousands of candidate "hits" dominated by anthropogenic RFI – has made machine learning an increasingly central tool for both signal detection and RFI rejection. Ma et al. (2023) presented the first large-scale deep-learning technosignature search, applying a $\beta$-convolutional variational autoencoder to 480 hours of Breakthrough Listen GBT data on 820 nearby stars in a semi-unsupervised anomaly-detection mode, recovering candidate signals missed by the standard TURBOSETI narrowband drift-search pipeline (Enriquez & Price 2019) and reducing the number of signals requiring human vetting by roughly two orders of magnitude, while returning eight signals of interest for re-observation. Building on this, Choza et al. (2024) identified and corrected a long-standing sensitivity mischaracterisation in TURBOSETI's signal-to-noise calibration, prompting a retrospective correction of quoted sensitivities in several earlier Breakthrough Listen publications (Lacki et al. 2021). More recently, unsupervised clustering methods (e.g. the "GLOBULAR" algorithm) have been developed specifically to separate genuine anomalies from RFI in narrowband technosignature searches without requiring labelled training examples (Jacobson-Bell et al. 2025), and further anomaly-detection searches have been applied to combined Parkes and GBT Breakthrough Listen archives (Poznanski et al. 2025). These efforts collectively demonstrate that machine-learning-based anomaly detection can recover classes of signal (broadband, non-linearly drifting, or otherwise morphologically unusual) that narrowband, linear-drift matched-filter searches such as TURBOSETI are, by design, insensitive to (Garrett et al. 2026), and that a purely classical, threshold-based search – of the kind we adopt for the present survey, following Mason et al. (2024) – necessarily leaves some fraction of the true technosignature parameter space unexamined. We return to this point in §6 as a natural direction for follow-up work once the present survey's baseline dataset exists.

### 2.5. *Statistical frameworks for interpreting non-detections*

A non-detection is not, by itself, a null result: it must be translated into a quantitative statement about which regions of transmitter parameter space have been excluded, and with what confidence. Wright et al. (2018) formalised this with the "Cosmic Haystack" framework, modelling the technosignature search space as an eight-dimensional volume (spanning, e.g., transmitter frequency, bandwidth, location, distance, and duty cycle) and showing that even the combined efforts of every major SETI programme conducted up to 2018 sampled a small fraction of any astrophysically plausible haystack volume – a result frequently paraphrased as showing that humanity has searched "the equivalent of a large hot tub's worth of the Earth's oceans". Sheikh (2020) subsequently proposed a complementary "nine axes of merit" rubric for comparing the relative scientific value of different technosignature search strategies (spanning axes such as detectability, transmitter cost and information content), providing a qualitative framework alongside Wright et al.'s quantitative one. Margot et al. (2023) developed an explicit formalism for converting a survey's non-detection rate into an upper bound on the fraction of stars in the search volume hosting a detectable transmitter, but noted explicitly that this inference is only valid if the observed sample is statistically representative of the underlying stellar population of the search volume – a requirement that most targeted, interest-selected samples (including our own earlier 100-star pilot list) do not, by design, satisfy. Most recently, Sheikh et al. (2025a) inverted the usual question, asking at what distance Earth's own technosignatures would themselves be detectable by an ALMA- or SKA-like observatory, providing a valuable calibration point (an "ichnoscale" of unity) against which the sensitivity of any given survey, including the present one, can be benchmarked.

### 2.6. *The case for a volume-limited, representative sample*

The great majority of the surveys reviewed above – including our own earlier 100-star ALMA pilot programme – select targets by some combination of proximity, Sun-like spectral type, known exoplanets, protoplanetary-disc or debris-disc detection, or membership of specially-constructed target lists such as the Earth Transit Zone. Such selections are entirely reasonable when the goal is to maximise the a priori probability of detecting a genuine technosignature per unit observing time, but they carry an unavoidable cost: they make it difficult to convert a survey's results into a statistically clean statement about the prevalence of technosignatures in the general stellar population, precisely because the sample is not representative of that population (Margot et al. 2023). A disc-hosting or exoplanet-hosting bias is particularly worth flagging in the ALMA context: ALMA-detectable circumstellar and protoplanetary discs are, almost without exception, orders of magnitude brighter than the present-day Solar System's own zodiacal dust cloud would appear at interstellar distances, so that samples selected *because* they have an ALMA-detected disc are biased towards young, dynamically active systems that are not necessarily more (and could plausibly be less) likely to host a mature, Earth-like technological civilisation than an old, quiescent, disc-poor star of the same spectral type.

The present survey is explicitly designed to avoid this selection effect. We construct our sample by identifying every star within 20 pc of the Sun that happens to fall within the field of view of at least one public ALMA observation – irrespective of whether that observation was proposed to study a disc, a stellar flare, an unrelated background source, or anything else – and irrespective of the target star's own spectral type, age, multiplicity, or known planet-hosting status. The star need not be at the pointing centre of the observation, provided it lies within the usable primary-beam field of view (see §3). This is, in effect, a volume-limited "bycatch" census of the ALMA archive: rather than sampling stars selected for astrophysical interest in the calibrator fields as in Mason et al. (2024), we sample every nearby star that ALMA happens to have observed for any purpose whatsoever, within a fixed physical volume. We recognise, and quantify explicitly in Paper II, that this approach is still subject to a residual, second-order selection effect – ALMA archival coverage of the immediate solar neighbourhood is itself not spatially uniform, since some regions and spectral types are more likely to have been proposed for ALMA time than others – but this bias operates on the presence of *any* ALMA coverage of a star, not on the astrophysical properties that are the actual subject of a technosignature search, and can therefore be characterised and reported as a well-defined completeness function of the 20 pc volume, rather than an implicit and unquantified bias in the target selection itself.

### 2.7. *Prospects for next-generation facilities*

The coming decade will see a substantial expansion of technosignature search capability. The Square Kilometre Array (SKA; Siemion et al. 2015; Tremblay et al. 2026), now under construction in Australia (SKA-Low) and South Africa (SKA-Mid), will combine wide instantaneous bandwidth, high angular resolution, and an architecture well suited to commensal, multi-beam technosignature searching, with projected sensitivity sufficient to detect Earth-analogue-level radar and communication leakage from significantly greater distances than is possible today. The next-generation Very Large Array (ngVLA; Ng et al. 2022) is specifically being designed with technosignature science as one of its commensal use cases, with proposed antenna configurations optimised to allow of order 64 simultaneous high-spectral-resolution commensal beams. Existing commensal systems at MeerKAT, the VLA (COSMIC), and the Murchison Widefield Array are already providing a preview of the observing modes these larger facilities will employ at scale. None of these facilities, however, offer ALMA's combination of angular resolution, spectral resolution and raw collecting area above $\sim 50$ GHz in the near term, so a systematic mm/submm technosignature census – of the kind begun here – is likely to remain a uniquely ALMA-enabled measurement for some years yet, and represents a natural complement to the lower-frequency, larger-sample surveys that will be enabled by SKA and ngVLA.

## 3. SAMPLE SELECTION

Our sample comprises every star within 20 pc of the Sun (based on Gaia DR3 parallaxes) with at least one public ALMA observation, of any band, science category, or proposal type, in which the star's true (proper-motion-corrected) position at the epoch of observation falls within the field of view of at least one execution block – not necessarily at the observation's pointing centre, since a substantial fraction of ALMA mosaics and wide single-pointing observations serendipitously cover more than one catalogued star. We deliberately do not weight target selection by the presence of a circumstellar disc, known exoplanets, spectral type, or any other astrophysical property beyond distance and archival data availability; disc presence and planet-hosting status are retained purely as descriptive metadata for the final sample, not as selection or ranking criteria. At the time of writing this identifies 120 unique target stars, spanning spectral types from early A (e.g. Sirius B, $\alpha$ CMa B) through to the latest, faintest M dwarfs (e.g. Proxima Centauri, TRAPPIST-1, Barnard's Star, Teegarden's Star), ranked by increasing distance from the Sun so that the nearest, most sensitively-constrained systems are processed first. The underlying pipeline is built to extend this volume-limited approach to 30, 40 or 50 pc as a natural next step, at the cost of a rapidly increasing target list and a correspondingly reduced fraction with existing ALMA coverage, since the archive's areal completeness at fixed exposure declines with distance.

## 4. DATA AND METHODOLOGY

For each target we mine the public ALMA science archive for all calibrated (or, where necessary, locally recalibrated from raw data) execution blocks covering the star's position, apply an efficiency-bounded download and processing budget per target to remain within the resource constraints of the shared compute facility used for this work, and search the resulting visibility data in two complementary ways: (i) a narrowband, Doppler-drift-corrected search for technosignature candidates, following the general methodology of Mason et al. (2024) and Enriquez et al. (2017), converting any non-detection into an equivalent isotropic radiated power (EIRP) limit via $\mathrm{EIRP}_{\min} = 4\pi d^2 S_{\min} \Delta\nu$; and (ii) a line-excluded continuum search for anomalous broad-band emission at the star's position, with an accompanying epoch-to-epoch variability check wherever multiple execution blocks exist. Stellar positions are propagated to the epoch of each observation using Gaia DR3 proper motions and parallaxes. The Doppler-drift search range applied to each target is set to the larger of (a) a generic frequency-scaled default following Mason et al. (2024)'s adoption of a literature fiducial with a threefold safety margin, and (b) where the target hosts one or more confirmed exoplanets, an explicit calculation of the maximum line-of-sight orbital acceleration (and hence maximum plausible transmitter drift rate) of the innermost known planet, so that a transmitter co-located with a known, short-period planet cannot be missed simply because its orbital motion drifts it out of an insufficiently wide search window.

### 4.1. *Closure-phase point-source vetting*

Any narrowband signal that crosses our detection threshold at the star's position is subjected to a further, automated test that is, to our knowledge, unavailable to any single-dish or phased-array-beamformed technosignature survey (Breakthrough Listen, FAST, the ATA, LOFAR, and similar all lack it, since it requires multiple independent baselines): the interferometric closure phase, the phase sum $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a closed triangle of baselines $i$-$j$-$k$. This quantity is a standard radio-interferometric identity that is theoretically *exactly zero* for an unresolved point source at *any* sky position – a point source's visibility phase is a pure linear function of $(u, v)$, which cancels identically around any closed baseline loop – and is, by construction, immune to per-station calibration errors. A candidate signal whose closure phase is statistically inconsistent with zero at the drift-tracked frequency of the hit is therefore evidence against it behaving as a genuine celestial point source, favouring instead a near-field, terrestrial, or instrumental origin. This test is independent of, and complementary to, the existing spatial-coincidence test (whether the on-source position shows a larger peak than the control positions): closure phase does not depend on where the source is, so it answers a different question – "does this behave like a coherent point source at all" – than the spatial test's "is it located at the star". We validated the closure- phase calculation against synthetic visibility data before applying it to any ALMA measurement, confirming zero closure phase for a noiseless simulated point source, statistical consistency with zero for a noisy simulated point source, and a large, clearly inconsistent scatter for a simulated non-point-source (RFI-like) signal. Because it is only triggered on an actual threshold-crossing candidate, this test adds negligible computational cost to the survey as a whole.

### 4.2. *Population-level, spectral-type-normalised continuum anomaly check*

Because our sample is volume-limited and not selected on disc-hosting or planet-host status, it supports a comparison that is not meaningful for the biased target lists used in essentially every prior technosignature survey: whether a star's measured continuum brightness is anomalous relative to the physically-expected flux of its own stellar photosphere, and relative to other stars of similar effective temperature already processed within the same survey. For each target we compute the expected photospheric flux density from the Planck function and the star's own solid angle, $S_\nu = \Omega \times 2h\nu^3/c^2 \times [\exp(h\nu/k_B T_{\mathrm{eff}}) - 1]^{-1}$ with $\Omega = \pi R_\star^2/d^2$, using Gaia DR3 photometric temperatures and, where available, Gaia GSP-Phot radii (supplemented by a coarse Pecaut & Mamajek main-sequence $T_{\mathrm{eff}}$–radius relation where a direct Gaia radius estimate is unavailable, as is the case for close to half of our sample – Gaia's astrophysical-parameter pipeline has known gaps for the very bright/nearby and very red/faint stars that a 20 pc sample is disproportionately composed of). Any large excess of the measured continuum flux (or, for a non-detection, its $5\sigma$ upper limit) above this photospheric baseline is flagged, and is additionally compared against the distribution of excess ratios already measured for other stars in the same coarse $T_{\mathrm{eff}}$ bin, once at least five such comparison stars exist, using a robust (median/MAD-based) outlier statistic. This check requires no additional raw-data retention beyond what the continuum measurement itself already produces, and updates incrementally as the survey proceeds.

## 5. RESULTS

As of this version of the paper, 19 of the planned 120 target/bands have been searched to completion (a further 2 completed their continuum measurement but not the narrowband search, owing to a data-quality failure in those specific execution blocks; both are flagged for an automatic, non-gating retry). The survey is strictly volume-ordered (nearest star first; §3), so the results reported here are weighted towards the very nearest stars in the sample, spanning 1.3–19.6 pc at the time of writing. Full per-target values, including the exact frequency range searched in each band,

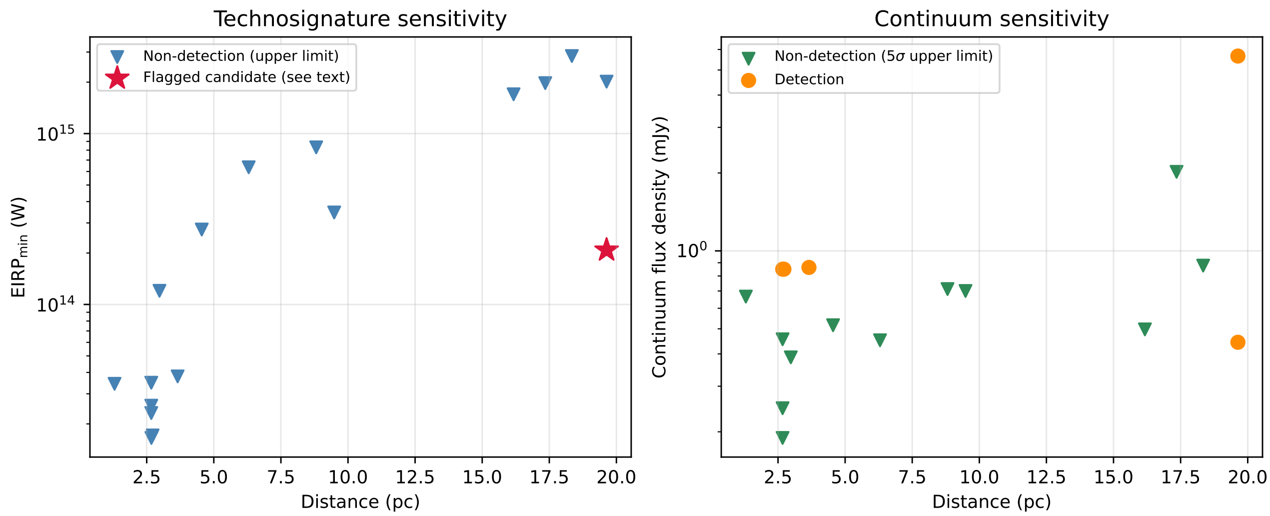


FIG. 1.— Technosignature EIRP upper limits (left) and continuum flux densities (right; downward triangles are $5\sigma$ upper limits, filled circles are detections) as a function of distance, for the 19 target/bands searched to date. The single automatically-flagged candidate ($\beta$ Pictoris, Band 3) is marked with a star and is identified in the text as CO(1–0) line contamination rather than a technosignature.

are given in Table 1 in Appendix A rather than in the main text; this section summarises the results and highlights points of interest.

### 5.1. *Technosignature search*

Seventeen execution blocks yielded a narrowband search result (two further targets, Barnard's Star and Wolf 359, had their continuum measurement succeed but their narrowband search fail outright, and carry no EIRP limit at this time). The resulting $\mathrm{EIRP}_{\mathrm{min}}$ limits span $1.6 \times 10^{13}$–$2.8 \times 10^{15}$ W, with a median of $2.1 \times 10^{14}$ W (Figure 1, left panel); as expected for a volume-ordered survey, the deepest limits are for the very nearest systems (e.g. $1.7 \times 10^{13}$ W for the UV Ceti binary at $\sim$2.7 pc). One narrowband hit was automatically flagged by the pipeline as a "credible technosignature candidate" (on-source peak signal-to-noise ratio above threshold and localised to the star's position, rather than appearing equally in the control positions): a signal towards $\beta$ Pictoris in ALMA Band 3 at 115.26 GHz. This frequency coincides, to within the spectral resolution of the observation, with the rest frequency of the CO($J = 1 \rightarrow 0$) rotational transition (115.271 GHz), and $\beta$ Pictoris is a well-known debris-disc system with previously reported CO emission; we therefore attribute this hit to line contamination rather than a genuine technosignature. No other candidate has survived both the spatial-coincidence and (where triggered) closure-phase vetting tests described in §4.1; in practice, the closure-phase test has not yet been exercised against a real ALMA measurement, since no other candidate has crossed the initial detection threshold to trigger it.

### 5.2. *Continuum search*

Of the 19 target/bands with a continuum measurement, five show a $\geq 5\sigma$ continuum detection and fourteen are non-detections (Figure 1, right panel). The non-detections reach $5\sigma$ upper limits of 0.19–3.6 mJy (median 0.59 mJy). Two of the five detections – catalogued separately in our target list as G 272-61A and G 272-61B – are in fact the two components of the well-known, closely separated visual binary UV Ceti, observed within a single ALMA pointing; the reported flux (0.85 mJy) is for the spatially unresolved, blended pair and cannot be attributed to either component individually. The remaining three detections ($\tau$ Ceti, and $\beta$ Pictoris in both Bands 3 and 6) are consistent with previously known circumstellar/debris material or stellar activity in these well-studied systems and are not treated as anomalous. The population-level anomaly check described in §4.2 has been applied to every target processed to date but has not yet flagged any star, and in most $T_{\mathrm{eff}}$ bins the survey does not yet have the minimum of five comparison stars needed for the population-relative component of the test; both the physical-baseline and population-relative statistics will accumulate statistical power as the survey proceeds.

## 6. DISCUSSION

The results presented here are necessarily preliminary: only 19 of 120 planned target/bands (16%) have been completed, and the survey is ordered to complete the nearest, most sensitively-constrained systems first, so the eventual full sample will extend these results to significantly larger distances and, correspondingly, shallower EIRP and continuum limits at the faint end. No credible technosignature has been found among the systems searched so far; the one automatically-flagged candidate is confidently attributable to CO(1–0) line emission rather than an artificial signal, underlining the importance – already highlighted by Mason et al. (2024) – of checking any millimetre-band narrowband hit against known molecular transitions before treating it as a candidate. Applying the statistical frameworks of Wright et al. (2018) and Margot et al. (2023) to derive a quantitative bound on the prevalence of detectable transmitters in the solar neighbourhood is deferred to a future version of this paper, once a statistically meaningful fraction of the

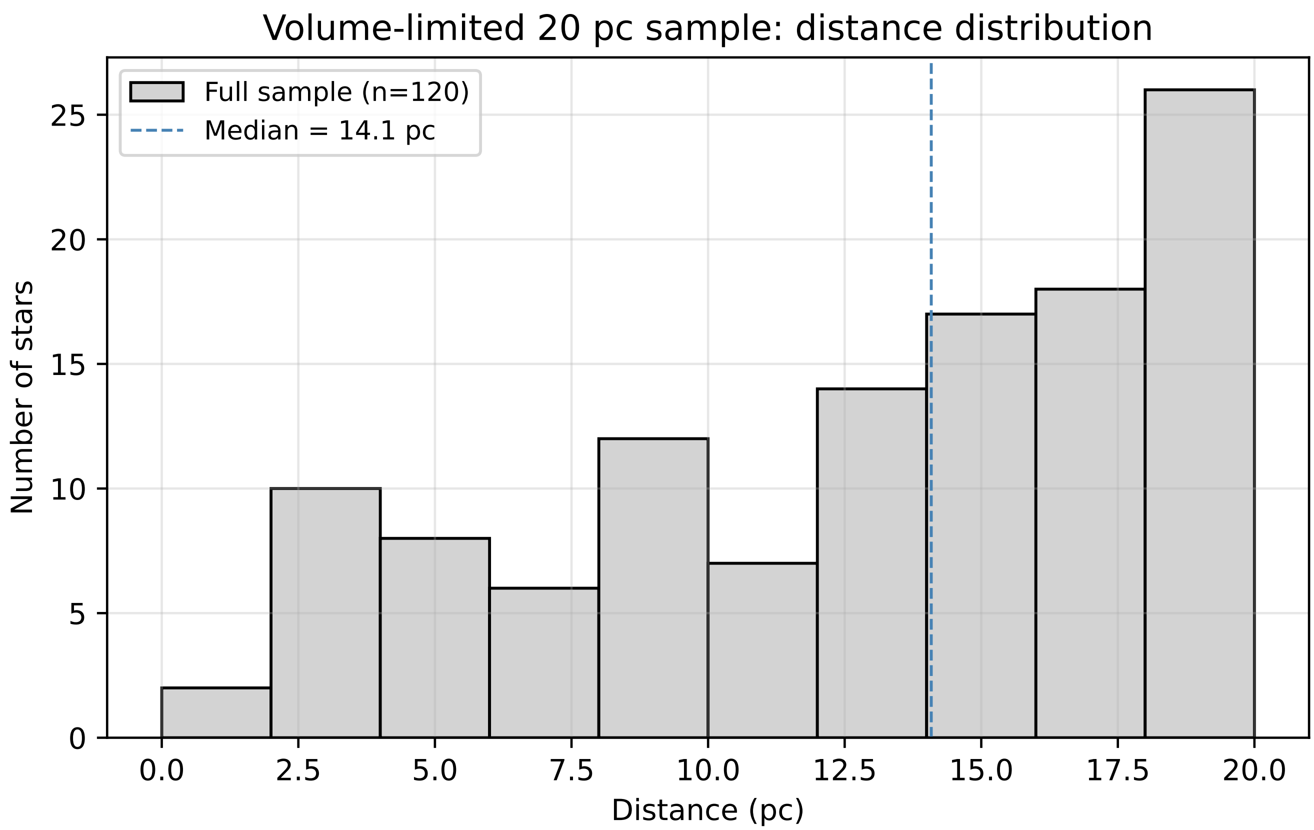


Fig. 2.— Distance distribution of the full 120-star volume-limited sample (median 14.1 pc). The rising number of stars per distance bin reflects the increasing shell volume with distance, moderated by the declining fraction of the local stellar population with existing public ALMA coverage.

volume-limited sample has been completed; with only 19 systems searched to date such a bound would not yet be usefully constraining.

The two supplementary analyses introduced in this work (§4.1, §4.2) are best understood as infrastructure investments rather than sources of results in their own right at this early stage. The closure-phase test has not yet been exercised against real data, simply because no candidate other than the already-explained $\beta$ Pictoris CO line has crossed threshold; it remains available, at negligible ongoing cost, for any future hit. The population-level anomaly check is accumulating the population of same-$T_{\rm eff}$-bin comparison stars it needs to become statistically powerful, and we expect it to become genuinely diagnostic only once several dozen targets per broad spectral-type bin have been processed. We regard both as a useful methodological contribution independent of whether they ultimately flag anything in this particular sample: to our knowledge, neither has previously been applied in a technosignature search, and both are made possible specifically by properties of this survey (interferometric data; a volume-limited, unbiased sample) that do not hold for the great majority of prior technosignature work reviewed in §2.

The UV Ceti unresolved-pair case (§5) illustrates a general caveat for any archival, position-agnostic survey of this kind: close visual binaries and multiple systems that are unresolved at the relevant ALMA angular resolution will necessarily share a single measurement, and any future statistical analysis of this sample's completeness must account for this rather than double-counting such pairs as independent trials.

## 7. CONCLUSIONS

We have described the design, methodology, and first results of a volume-limited technosignature search of all 120 stars within 20 pc with public ALMA archival coverage, irrespective of disc-hosting or planet-host status. Nineteen of the planned 120 target/bands have been completed as of this version, yielding EIRP limits of $1.6 \times 10^{13}$–$2.8 \times 10^{15}$ W and fourteen continuum non-detections reaching a median $5\sigma$ limit of 0.59 mJy, with no credible technosignature identified. We have introduced two analyses – interferometric closure-phase vetting and a population-level, spectral-type-normalised continuum anomaly check – that exploit properties unique to this survey and are, to our knowledge, new to the technosignature literature; both are now running as a standard part of the pipeline and will accumulate statistical power as the survey proceeds. This paper will be updated as further targets are completed, and we anticipate extending the volume limit to 30, 40, and 50 pc as a natural next step once the 20 pc sample is complete.

ACKNOWLEDGEMENTS

TABLE 1
PER-TARGET/BAND RESULTS, ORDERED BY DISTANCE. FREQUENCY RANGE IS IN GHz; $\mathrm{EIRP_{min}}$ IS IN W. "CAND." INDICATES WHETHER THE AUTOMATED PIPELINE FLAGGED THE NARROWBAND SEARCH AS A CREDIBLE TECHNOSIGNATURE CANDIDATE (Y/N); SEE §5 FOR THE DISPOSITION OF THE ONE Y IN THIS TABLE.

| Target | Band | Dist. (pc) | Freq. range (GHz) | $\mathrm{EIRP_{min}}$ (W) | Cand. | Continuum (mJy) | Notes |
|---|---|---|---|---|---|---|---|
| Proxima Cen | B6 | 1.30 | 223.09–224.91 | 3.43e+13 | N | 0.666 (UL) | |
| NAME Barnard's star | B7 | 1.83 | ... | ... | ... | 3.622 (UL) | narrowband search failed |
| Wolf 359 | B7 | 2.41 | ... | ... | ... | 3.380 (UL) | narrowband search failed |
| Sirius B | B3 | 2.67 | 92.13–93.86 | 2.56e+13 | N | 0.189 (UL) | |
| Sirius B | B4 | 2.67 | 142.13–143.86 | 2.31e+13 | N | 0.247 (UL) | |
| Sirius B | B5 | 2.67 | 195.13–196.86 | 3.48e+13 | N | 0.455 (UL) | |
| G 272-61B | B3 | 2.67 | 89.63–91.36 | 1.65e+13 | N | 0.850 (det.) | unresolved UV Cet AB pair, blended flux |
| G 272-61A | B3 | 2.72 | 89.63–91.36 | 1.71e+13 | N | 0.850 (det.) | unresolved UV Cet AB pair, blended flux |
| CD-23 14742 | B6 | 2.98 | 223.09–224.91 | 1.20e+14 | N | 0.388 (UL) | |
| tau Cet | | 3.65 | 229.63–231.47 | 3.79e+13 | N | 0.863 (det.) | |
| GJ 674 | B6 | 4.55 | 223.09–224.91 | 2.74e+14 | N | 0.516 (UL) | |
| GJ 581 | | 6.30 | 223.09–224.91 | 6.35e+14 | N | 0.451 (UL) | |
| GJ 849 | | 8.81 | 229.08–230.92 | 8.30e+14 | N | 0.712 (UL) | |
| HD 285968 | | 9.49 | 229.04–230.88 | 3.45e+14 | N | 0.700 (UL) | |
| HD 23484 | B6 | 16.17 | 229.61–231.45 | 1.70e+15 | N | 0.497 (UL) | |
| HD 10647 | B7 | 17.35 | 330.35–332.19 | 1.97e+15 | N | 2.020 (UL) | |
| HD 53143 | | 18.34 | 229.60–231.44 | 2.84e+15 | N | 0.877 (UL) | |
| bet Pic | B3 | 19.63 | 114.83–115.69 | 2.08e+14 | Y | 0.444 (det.) | CO(1-0) line, see text |
| bet Pic | B6 | 19.63 | 230.08–231.00 | 2.01e+15 | N | 5.659 (det.) | |

This paper makes use of the following ALMA data: [the full list of ALMA project codes used will be compiled once the survey is complete; see Table 1 in Appendix A for the individual execution blocks searched to date]. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC).

## DATA AVAILABILITY

The reduced data products underlying this paper, together with the full target-selection and analysis pipeline, are being made available at `https://github.com/Tilanthi/SETI` as the survey progresses. Raw ALMA visibility data are publicly available from the ALMA Science Archive at `https://almascience.org`.

## APPENDIX

## FULL PER-TARGET RESULTS TABLE

Table 1 lists the full per-target/band results underlying the summary given in Section 5: the narrowband technosignature frequency range actually searched (not the full ALMA tuning – only the single finest-resolution spectral window is searched, per §4), the resulting $\mathrm{EIRP_{min}}$ limit, whether the automated pipeline flagged a credible candidate, and the continuum flux density (or, for non-detections, the $5\sigma$ upper limit, marked UL). This table will grow as the survey proceeds; the version current at the time of writing is reproduced here so that the summary statistics quoted in the main text are fully traceable.

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